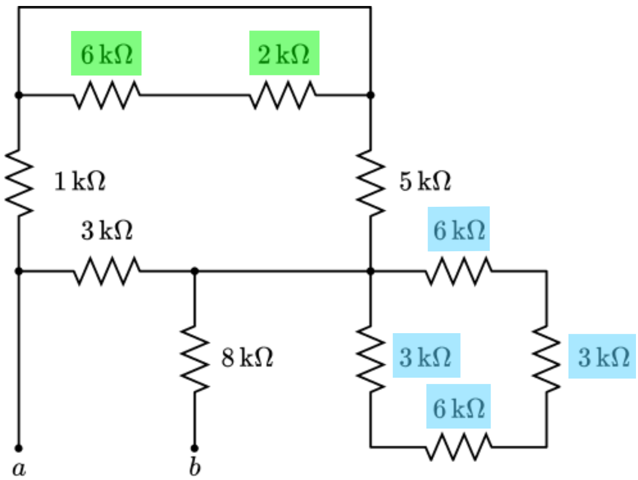
Practice Quiz Two Solution: Equivalent Resistance (Easy)

For the circuit shown below, find the equivalent resistance R_{ab}.

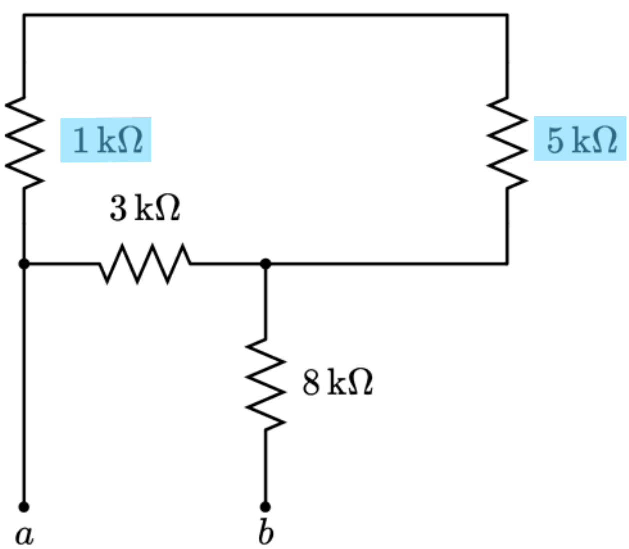
Step One: Identify short and open resistors



If we combined each highlighted resistor in series, the equivalent resistor would be shorted, or connected to one node on both sides.

The resistor attached to the open terminal pair is not considered open, since we are solving with respect to that pair.

Step Two: Combine Resistors



$$(1k\Omega + 5k\Omega) \parallel 3k\Omega = 2k\Omega$$

$$2k\Omega + 8k\Omega = 10k\Omega$$

$$R_{ab} = 10k\Omega$$

Self-Grading Rubric	
Identify 1st short	+2 points
Identify 2nd short	+2 points
Identify 1kΩ is in series with 5kΩ	+2 points
Identify two parallel resistors	+2 points
Correct final answer	+2 points

giannacco ecedu.org